PPP-F-00690a(ARMY-GL)
October 3, 1967

Int. Revision of
PPP-F-690
April 2, 1964

INTERIM FEDERAL SPECIFICATION

FROZEN FRUITS, CONCENTRATED FRUIT JUICES, AND VEGETABLES; PACKAGING, PACKING, LABELING, AND MARKING OF

This interim Federal specification was developed by the U.S. Army Natick Laboratories, Natick, Massachusetts based upon currently available technical information. It is recommended that Federal Agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this interim Federal specification as a valid exception to Federal Specification PPP-F-690.

1. SCOPE

1.1 This specification covers the packaging, packing, labeling, and marking requirements for the domestic and overseas shipment of frozen fruits, concentrated fruit juices, and vegetables.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

FEDERAL SPECIFICATIONS

PPP-B-636 - Box, Fiberboard.

PPP-B-566 - Boxes, Folding, Paperboard.

FEDERAL STANDARDS

Fed. Std. No. 123 - Marking for Domestic shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index

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of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

MILITARY STANDARDS

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. The following publications form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

GOVERNMENT

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

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(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

USDA-CMS Visual Aid #1, Visual Aids for Inspection of Metal Containers.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 Materials. All packaging and packing materials shall be of good commercial quality and as herein specified, and shall conform to the provisions of the Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

3.2 General. Frozen fruit, concentrated fruit juices and vegetables shall be prepared for shipment in accordance with the requirements of this specification. When the product specification requirements are in conflict with the requirements of this specification, the requirements in the specification of latest date shall take precedence.

3.3 Packaging. The product shall be packaged in accordance with level B or level C, as specified (see 6.1).

3.3.1 Level B. The product shall be packaged in accordance with 3.3.1.1, 3.3.1.2, or 3.3.1.3, as applicable.

3.3.1.1 Frozen fruits. Not more than 10 pounds of product shall be packaged in an open top style, round, metal can with soldered side seam and compound-lined, double-seamed ends. The can shall be made throughout with tin plate in the amount specified by the can manufacturer for the specific product being packaged. The can shall be hermetically sealed and shall not leak when tested in accordance with 4.4.1.

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3.3.1.2 Frozen concentrated fruit juices. The product shall be packaged in the size can specified (see 6.1) and in accordance with the fill requirements of table I. The can shall be an open top style, round, metal can with soldered side seam and compound-lined, double-seamed ends. The can shall be made throughout from 0.10 pound electrolytic tinplate per base box. The can shall be hermetically sealed and shall not leak when tested in accordance with 4.4.1.

TABLE I - Can sizes, fill, and arrangement in shipping container for frozen concentrated fruit juices

Can size	Volume of Contents	No. in Container	Arrangement		
			Length	Width	Depth
202 x 314	6 ounces	48	6	4	2
211 x 414	12 ounces	48	6	4	2
401 x 411	1 pint 10 ounces	24	4	3	2
401 x 509-10	1 quart	12	4	3	1
404 x 414	1 quart	12	4	3	1
404 x 500	1 quart	12	4	3	1
404 x 700	1 quart 14 ounces	12	4	3	1
603 x 700	3 quarts	6	3	2	1

3.3.1.3 Frozen vegetables. Not more than 5 pounds of the product shall be packaged in accordance with 3.3.1.3.1 or 3.3.1.3.2.

3.3.1.3.1 The product shall be packaged in cartons, the size, style, and net contents of which shall be the same as that normally used by the packer. The carton shall be over-wrapped and completely heat sealed in one of the materials specified

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below. Alternatively, the product may be inclosed in a heat sealable bag made from one of the materials specified below and the sealed bag inclosed in a carton. If a bag is used, the carton need not be overwrapped. The overwrapping or bag materials shall be plastic film or waxed paper of the types normally used in packaging frozen foods and meeting the requirements of Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

3.3.1.3.2 Alternatively, the carton may be a seal-end carton in accordance with style I, type A, class a of PPP-B-566 fabricated from minimum 0.014 inch thick solid bleached sulphate with minimum 0.005 inch thick medium density polyethylene (treated for gluing) extruded to the inside of the board with an overprint varnish (or polyethylene) applied to the outside of the board. The carton shall be closed with a hot melt adhesive, applied so that the outside flap is thoroughly bonded to the opposite flap of the carton. The adhesive shall not fail when subjected to a temperature of minus 20° F. for a minimum of 2 hours (see 4.4.2).

3.3.2 Level C. The product shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets these requirements.

3.4 Packing. The product shall be packed in accordance with level B or C, as specified (see 6.1).

3.4.1 Level B. Not more than 60 pounds of fruit product, nor more than 40 pounds of vegetable product, or the concentrated juices in the number and arrangement specified in table I shall be packed into a snug-fitting fiberboard box constructed and closed in accordance with style RSC, type CF or SF, class domestic, method II closure, of PPP-B-636.

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3.4.1.1 When specified (see 6.1 and 6.2) the shipping container shall be constructed, closed, and reinforced in accordance with style RSC, V3c or V3s, of PPP-B-636.

3.4.2 Level C. The product shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The shipping container shall be in accordance with rules or regulations applicable to the mode of transportation.

3.5 Labeling and marking.

3.5.1 Civil agencies.

3.5.1.1 Unit containers. Any commercial labeling and additional labeling as specified that complies with the Federal Food, Drug, and Cosmetic Act, and Regulations Promulgated Thereunder, is acceptable.

3.5.1.2 Cooking instructions (vegetable items only). Unless otherwise specified, each carton or package shall include cooking instructions, on the outer wrapper, or on an attached label.

3.5.1.3 Shipping containers. Shipping containers shall be marked in accordance with the requirements of Fed. Std. No. 123, and such other information as specified by individual agencies.

3.5.2 Military agencies.

3.5.2.1 Unit containers. Unless otherwise specified, labeling as specified in 3.5.1.1 and 3.5.1.2 will be acceptable.

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3.5.2.2 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129 and as specified in 3.5.2.3.

3.5.2.3 Additional markings. In addition to the markings specified in 3.5.2.2 the following precautionary markings shall appear on the top of the container, in letters between 1 and 1-1/2 inches high, all letters being of the same height.

KEEP FROZEN

(0°F. or below.)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated herein.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3 Inspection of the end item.

4.3.1 Examination of the end item. Examination of the end item shall be in accordance with the classification of defects, inspection levels and acceptable

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quality levels expressed in defects per hundred units as set forth below. The lot shall be expressed in units of shipping containers.

Examination	Sample unit	Inspection level	AQL'S			
			Major A	Major B	Minor	Total
Table II	One filled primary container	S-4	1.5	2.5	--	6.5
Table III	One filled primary container	S-2	--	1.5	2.5	--
Table IV	One filled primary container	S-3	--	2.5	--	--
Table V	One filled primary container	S-3	--	2.5	--	--

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TABLE II - Examination of packaging[1]

Categories			Defects	Cans	Cartons 0/wrap[5]	Cartons Bags[6]
Major A:	Major B:	Minor:				
101			Leaker	X		
	151		False seam	X		
	152		Improperly closed	X		
	153		Not type specified	X		X
	154		Severe dent	X		
		201	Moderate dent	X		
	155		Dent causing sharp ridge	X		
	156		Pitted rust[2]	X		
	157		Nomenclature missing, incorrect or illegible.	X	X	
		202	Labeling information other than nomenclature missing incorrect or illegible	X	X	
		203	Not clean[3]	X	X	X
		204	Excessive amount of flux on ends or sides[4]	X		
102			Not completely heat sealed or bonded		X	X
103			Missing		X	X
	158		Not made of specified materials		X	X
		205	Bulging ends 3/16-inch to 1/4-inch beyond lip	X		
	159		Bulging ends more than 1/4-inch beyond lip	X		

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[1]Classification of can defects shall be in accordance with USDA-CMS Visual Aid #1, Visual Aids for Inspection of Metal Containers.

[2]Rust that can be removed by wiping with a soft cloth will not be considered as pitted rust.

[3]Cans showing a very thin film of grease which is discernible to the touch but not readily discernible visually are considered to be clean.

[4]Traces of flux at edges of soldered areas, normally encountered in can manufacturing but not enough to cause corrosion, will not be considered excessive.

[5]Applicable when overwrap is used.

[6]Applicable when bag is used.

TABLE III - Examination for net weight of packages[1]

Categories		Defects
Major B:	Minor	
	201	Net weight more than 2 percent under net weight marked on package. (Report net weight to nearest 1/4 ounce.)
	151	Net weight exceeds the amount specified in table I or II for level B packaging.

[1]The lot shall be rejected if the sample data indicates a lot average net weight less than net weight marking on packages.

TABLE IV - Examination for leakage of cans

Categories	Defects
Major B:	
151	Steady progression of bubbles.[1]

[1]When tested in accordance with 4.4.1

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4.3.1.1 Examination of shipping container for net weight and marking. Examination of the filled and closed shipping container for net weight and marking shall be performed to determine conformance to the applicable requirements in section 3. Weighing of shipping containers may be performed concurrently with packing and marking of containers or performed prior to shipment. Classification of defects found during examination shall be in accordance with table V.. The sample unit shall be one filled and marked shipping container. The lot shall be expressed in terms of shipping containers. The sample size for examination shall be the number of sample units indicated by inspection level S-3. The acceptable quality level (AQL) expressed in defects per hundred units, shall be 2.5 for major defects and 6.5 for total defects.

TABLE V - Examination for net weight and marking

Category		Defects
Major	Minor	
	201	Net weight more than specified.
151		Nomenclature missing incorrect or illegible.
152		Precautionary marking omitted, or not as specified (see 3.5.2.3).
	202	Other markings omitted, incorrect, illegible, or not as specified (see 3.5.2.2).

4.3.1.2 Condition inspection of containers. When inspection of product for quality and condition is performed by the U.S. Department of Agriculture, the United States Standards for Condition of Food Containers pertaining to processed fruits and vegetables shall apply.

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4.3.1.3 Examination of shipping container. When containers are required to be in accordance with PPP-B-636, examination of the filled and closed shipping container shall be in accordance with that specification.

4.4 Tests.

4.4.1 Leakage test. The filled and sealed can of thawed product shall be tested by submerging /in water, contained in a desiccator or other suitable container, and maintaining a vacuum of ten inches of mercury (atmospheric pressure 29.9 inches) for at least 30 seconds. A leak consists of a steady progression of bubbles. Isolated bubbles caused by entrapped air are not considered signs of leaks.

4.4.2 Temperature test. The sealed carton (see 3.3.1.3.2) shall be tested for separation at glue joint by placing in a cold box with a temperature of minus 20° ± 2° F. for a period of 2 hours.

TABLE VI. Temperature test

Categories	Defects
Major B	
151	Any flap separation from carton at glue joint[1].

[1]When tested in accordance with 4.4.2.

4.5 Standby test samples. The Government reserves the right to withdraw and hold standby test samples of components or finished product or both (quantity of which shall not be more than twice that required by this specification) for inspection purposes. These samples, if not used, will be returned to the vendor.

5. PREPARATION FOR DELIVERY - Not applicable.

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6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Size and kind of packaging required.

(c) Selection of applicable levels of packaging and packing (see 3.3 and 3.4).

(d) When packing specified in 3.4.1.1 is required.

6.2 Packing specified in 3.4.1.1 is intended to provide the limited Military protection required for shipment of supplies to certain specified overseas destinations, and transfer at sea.

Military custodians:

Army - GL

Preparing activity:

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION Frozen Fruits, Concentrated Fruit Juices, and Vegetables; Packaging, Packing, Labeling, and Marking Of - PPP-F-00690a(Army-GL)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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